

# KavachAI Plant Investigation & Inspection Report

Sovereign Industrial Agentic AI Workbench | Formal Plant Engineering Record

| Investigation ID                     | Timestamp / Date | Confidence / Verification | Overall Status     |
|--------------------------------------|------------------|---------------------------|--------------------|
| 750a889d-eb9b-463d-b203-2bc672a95c73 | 2026-09-08       | 95%   VERIFIED            | ATTENTION REQUIRED |

Investigation Query: "Investigate about P-102"

P&ID Process Flow: T-101 -> P-102 -> V-204 -> R-101

## Key Findings & Grounded Evidence

|                                                                                                                                                                                                                                                                           |             |
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| <p><b>1. Increasing Operating Temperature</b></p> <p>The temperature of Pump P-102 has risen from 68 °C in January 2026 to 71 °C in April 2026, indicating a significant increase in operating temperature.</p> <p>Citations: P&amp;ID; Drawing • Document, p. 1</p>      | [SUPPORTED] |
| <p><b>2. Vibration Level Fluctuation</b></p> <p>The vibration level of Pump P-102 has fluctuated from 2.1 mm/s in January 2026 to 2.8 mm/s in April 2026, suggesting potential issues with the pump's operation.</p> <p>Citations: P&amp;ID; Drawing • Document, p. 1</p> | [SUPPORTED] |

## Synthesized Conclusion & Engineering Action Plan

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| <p><b>Engineering Synthesis:</b></p> <p>The available evidence indicates a condition change over the observed period. Engineering inspection is recommended. The evidence does NOT establish imminent failure.</p> <p><b>Operational Recommendation:</b></p> <ol style="list-style-type: none"><li>Schedule laser shaft alignment and coupling inspection during the nearest maintenance window.</li><li>Conduct high-frequency spectral vibration analysis to track bearing defect frequency progression.</li><li>Perform lubrication oil analysis on drive-end bearings.</li></ol> |
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| <p><b>Automated Verification:</b> PASSED (95% Grounding)</p> <p><b>Inference Host:</b> Sovereign Local Engine (Air-Gapped)</p> | <p><b>Forensic Integrity:</b> Append-Only SQLite Audit Trail</p> <p><b>Sovereignty:</b> Zero Cloud Egress / Confidential Operations</p> |
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